

Rev	Date	Eng	Changes
AA	2026-10-07	NB	First release, derived from Sofar mote 000639-AB

nereus_Pi_shield_ADIN2111

Variant: Standard

P/N: PCB-000001-AA

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- 2 Top-Level
- 3 ADIN2111 (Single-Pair Ethernet)
- 4 Load Switch (payload port)
- 5 PoDL (Power over Data Line)
- 6 Bus Power Monitor
- 7 Power (buck converters)

Pi shield renders: TBD (after layout)

I2c Address Table

Device	I2C Address
Power Monitor	0x41

DIMENSIONS: TBD (after layout)

Top-Level Schematic
File: BM Mote 1 Master.kicad_sch

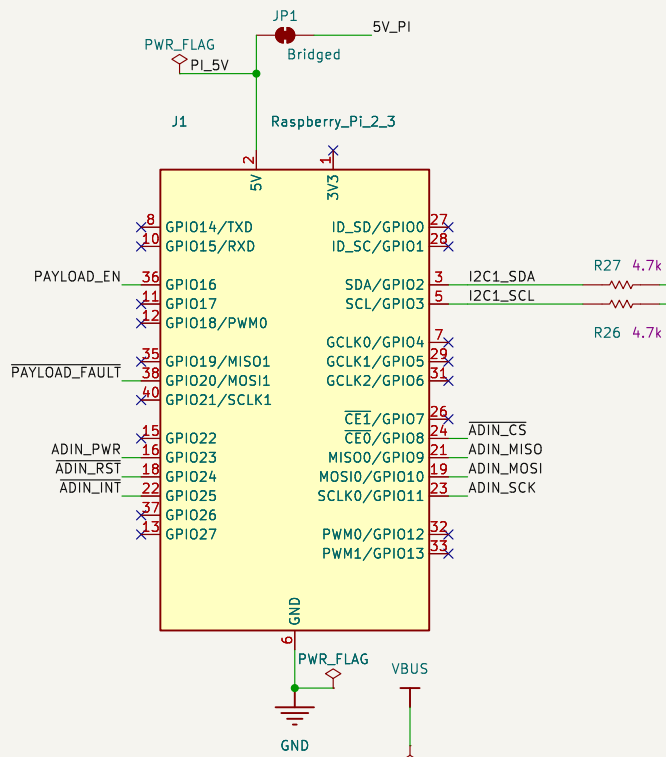


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<https://creativecommons.org/licenses/by-nd/4.0/legalcode>

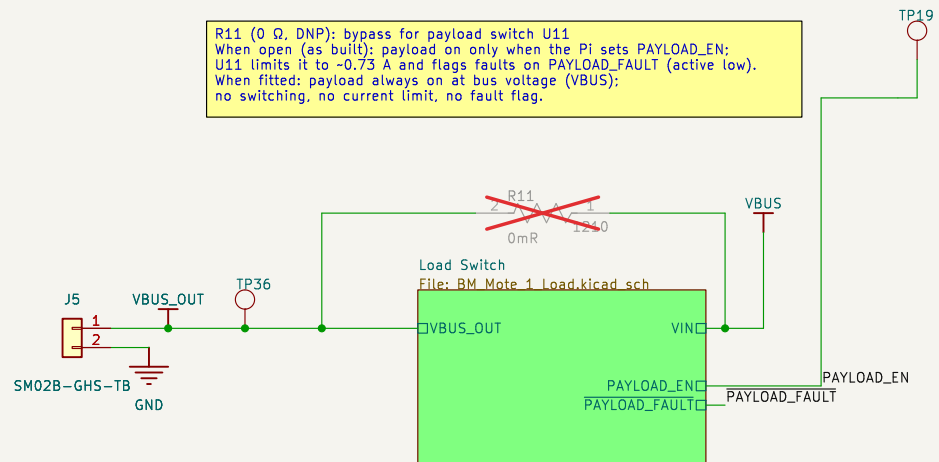


Derived from Sofar Bristlemouth mote 000639-AB Author: Nick Buemond nereus_Pi_shield_ADIN2111 · PCB-000001-AA			
Sheet: / File: nereus_Pi_shield_ADIN2111.kicad_sch			
Title: Cover Page			
Size: A3	Date: 2026-10-07		Rev: AA
KiCad E.D.A. 9.0.6		Id: 1/7	

JP1 bridged as built: the shield powers the PI (D14).
Bridged: never plug a 5 V source into the PI's USB (charger, PC, self-powered hub), shield powered or not.
USB peripherals powered by the PI are OK. PI powered elsewhere (battery pack): cut the bridge before potting;
refit a 0R 1210 (CRCW121000020EA) to reconnect.

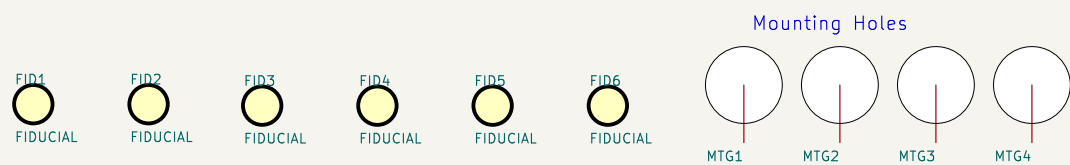


RASPBERRY PI – 40–PIN HEADER

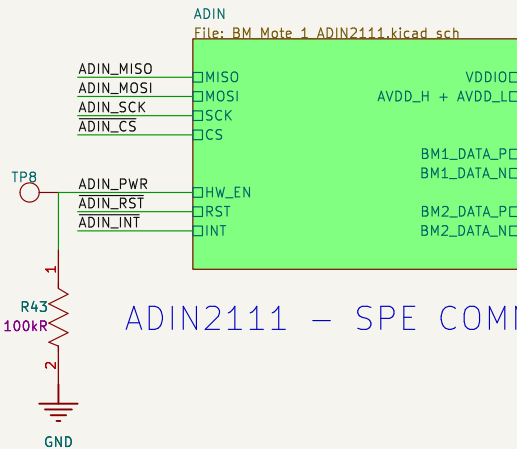


Payload power out: J5, JST GH 1.25 mm (023).
Pin 1 VBUS_OUT: 16–30 V bus, switched by U11.
–0.73 A limit. Pin 2 GND.
Mate the cable before potting.

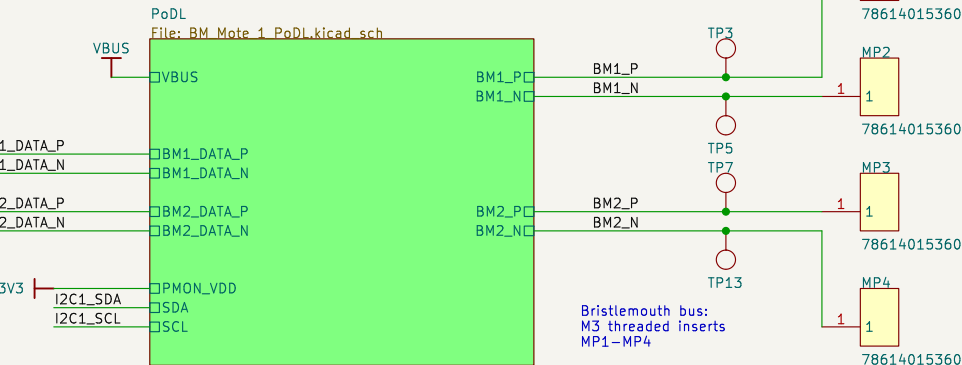
LOAD SWITCH – PAYLOAD POWER



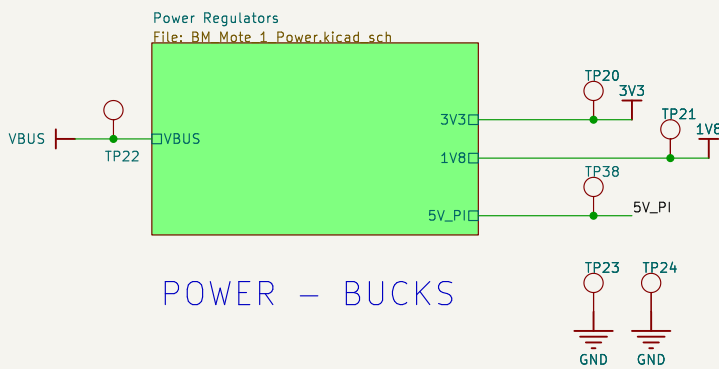
Shield logic runs off 3V3;
ADIN AVDD runs off 1V8.



ADIN2111 – SPE COMMS



PoDL – POWER OVER DATA LINE



POWER – BUCKS

Derived from Sofar Bristlemouth mote 000639–AB
Author: Nick Buemond
nereus_PI_shield_ADIN2111 · PCB–000001–AA

Sheet: /Top–Level Schematic/
File: BM_Mote_1_Master.kicad_sch

Title: Top–Level

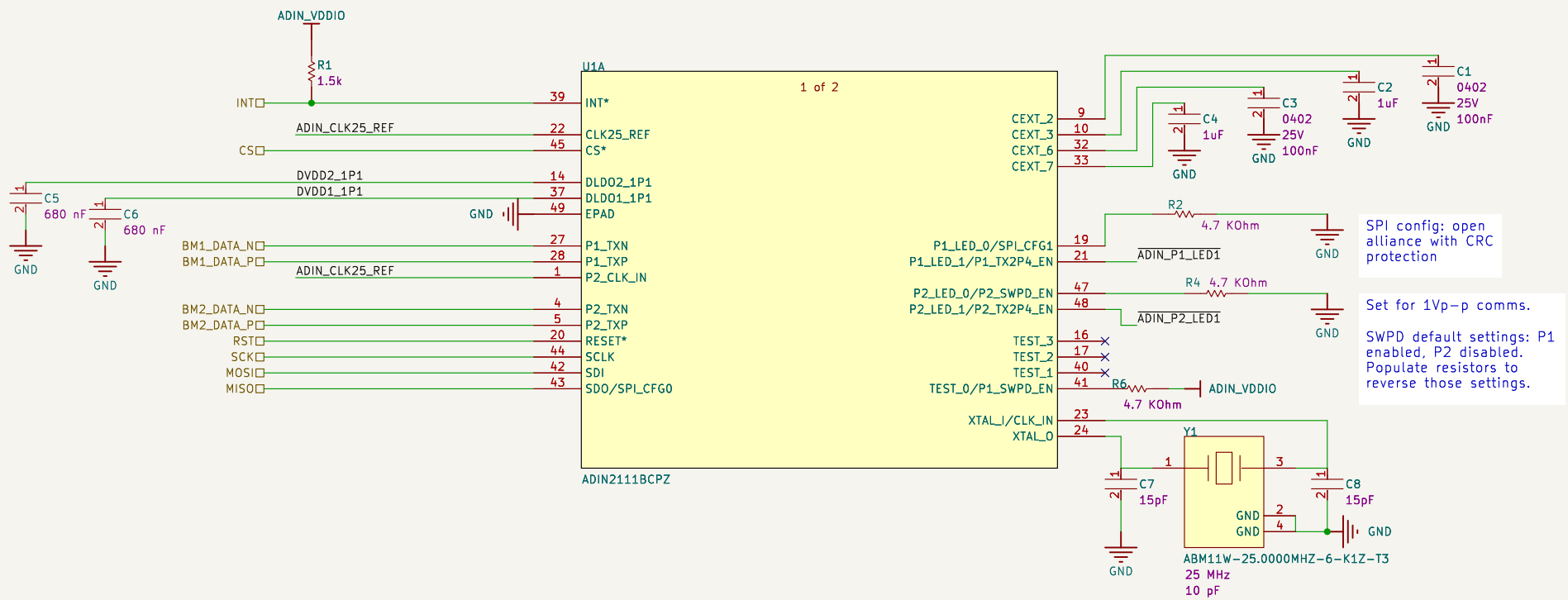
Size: A2 Date: 2026–10–07

Rev: AA

KiCad E.D.A. 9.0.6

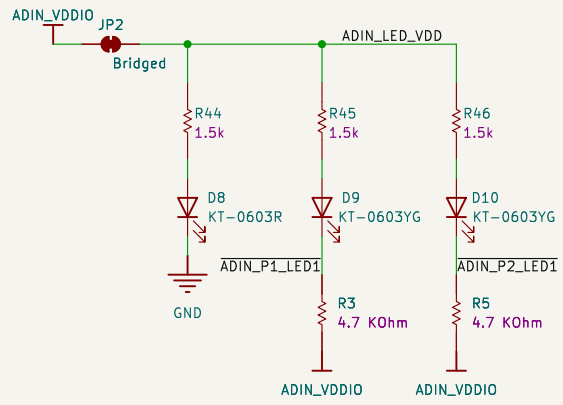
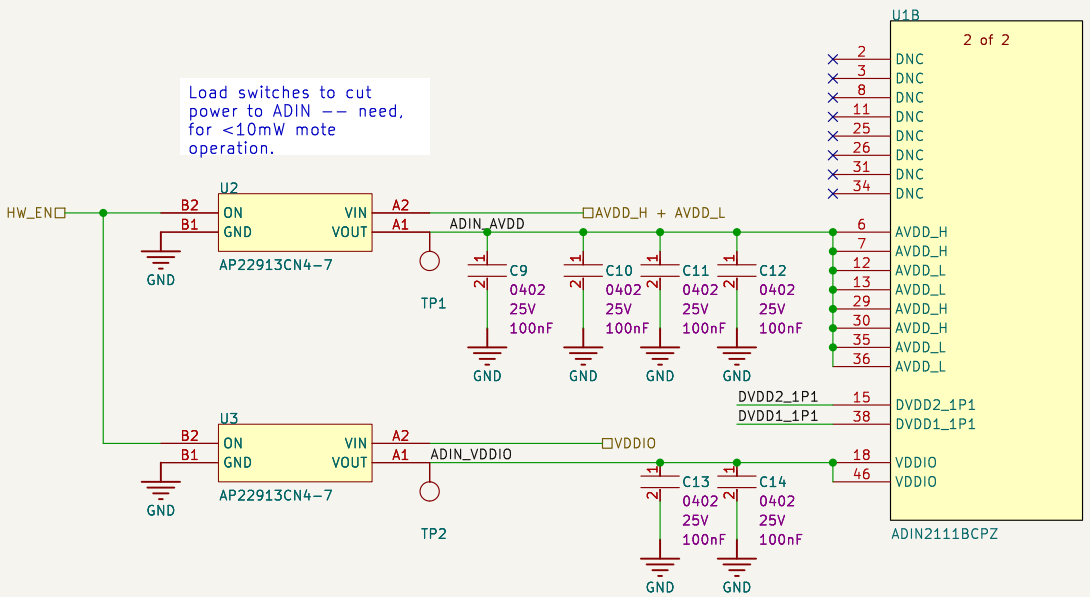
Id: 2/7

ADIN2111 Single-Pair Ethernet



ADIN2111 status LEDs: power, port 1 / port 2 link-activity

ADIN2111 supplies: U2 (AVDD) and U3 (VDDIO) switched on by ADIN_PWR

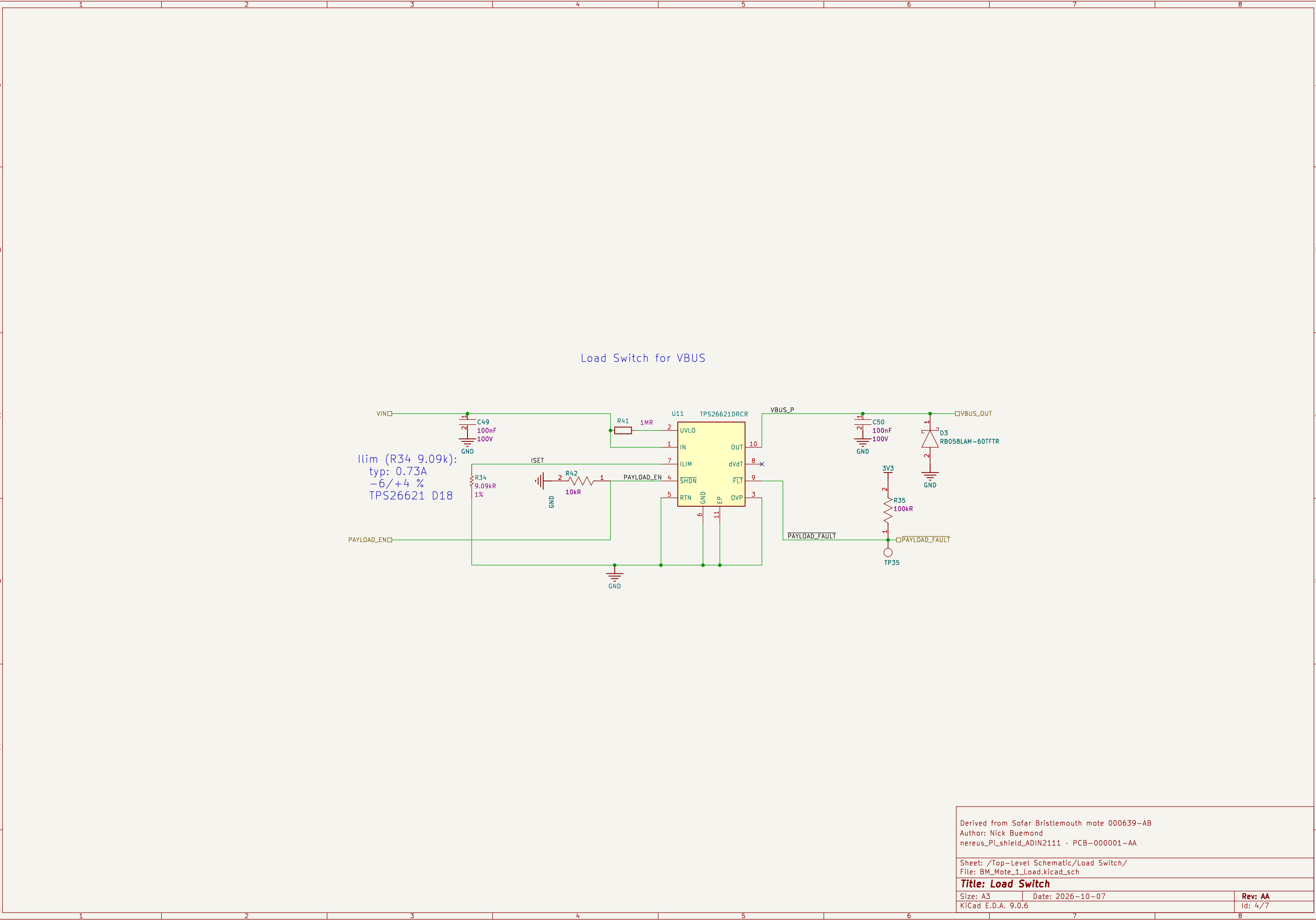


ADIN status LEDs (D25): D8 red = ADIN powered (ADIN_VDDIO on). D9 / D10 yellow-green = port 1 / port 2 link/activity. ADIN pins 21 / 48, active low. Off until the Pi enables each port's LED_1 over SPI; port 2 is also asleep until woken. R3 / R5 keep the TX2P4_EN straps high (1.0 V p-p). Cut JP2 to disable all three LEDs.

Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

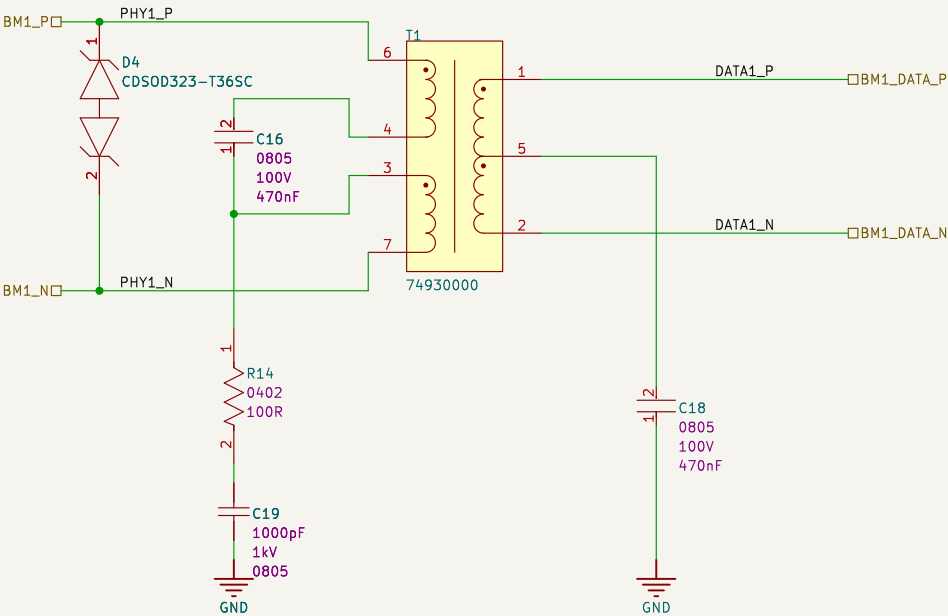
Sheet: /Top-Level Schematic/ADIN/
File: BM_Mote_1_ADIN2111.kicad_sch

Title: ADIN2111		
Size: A3	Date: 2026-10-07	Rev: AA
KiCad E.D.A. 9.0.6		Id: 3/7

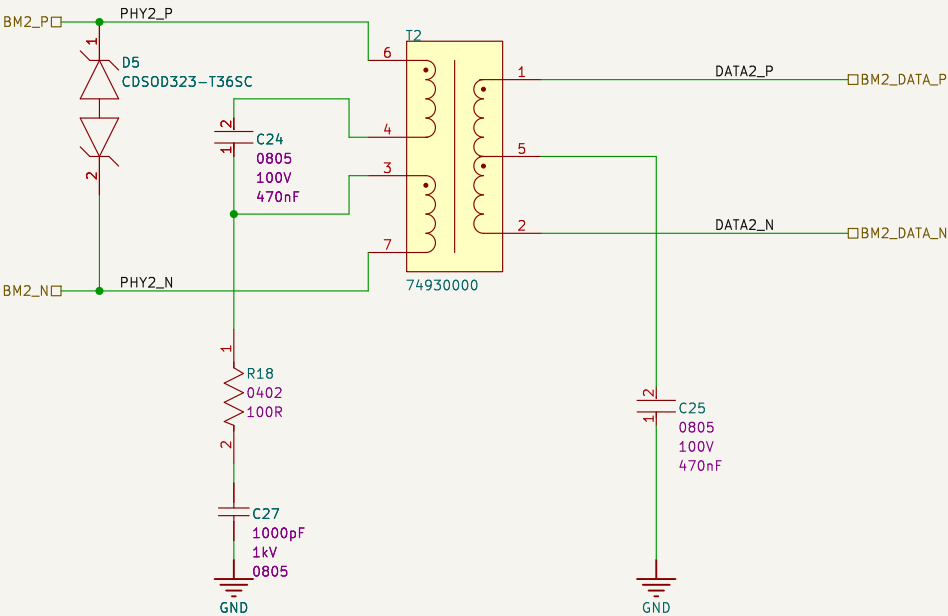


Power Over Dataline

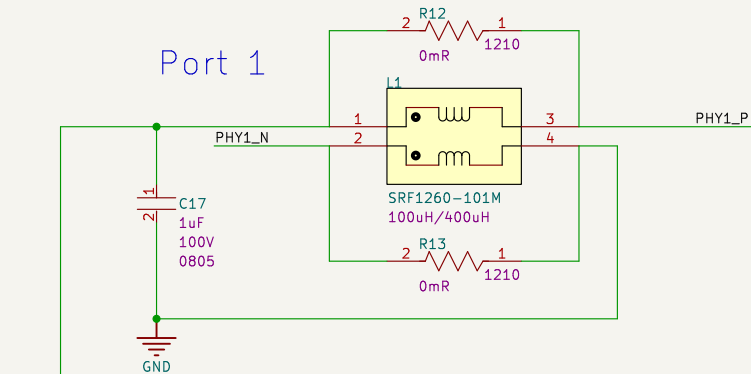
Coupling Network – Port 1



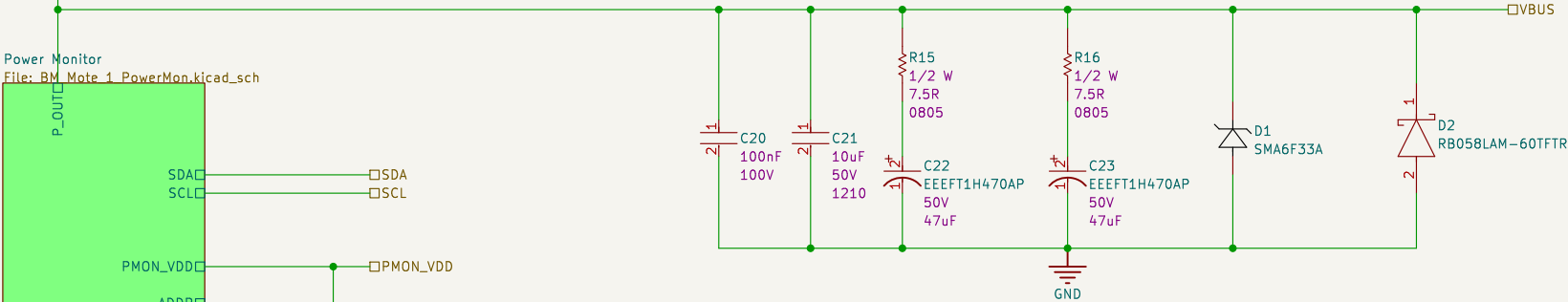
Coupling Network – Port 2



Port 1

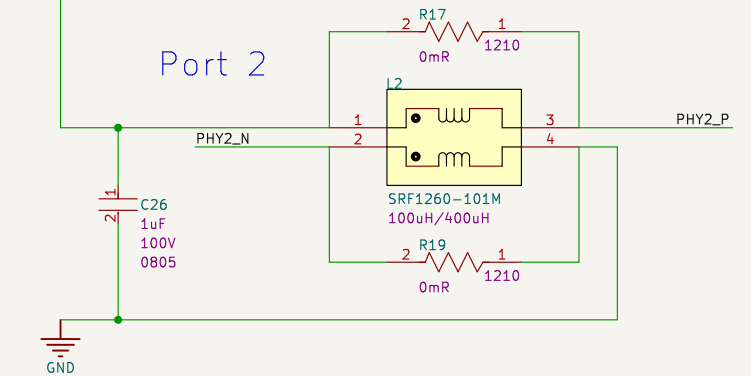


DC Connection + Circuit protections



Snubber resistors must be at least 1/2 W, to support hot-plugging.

Port 2



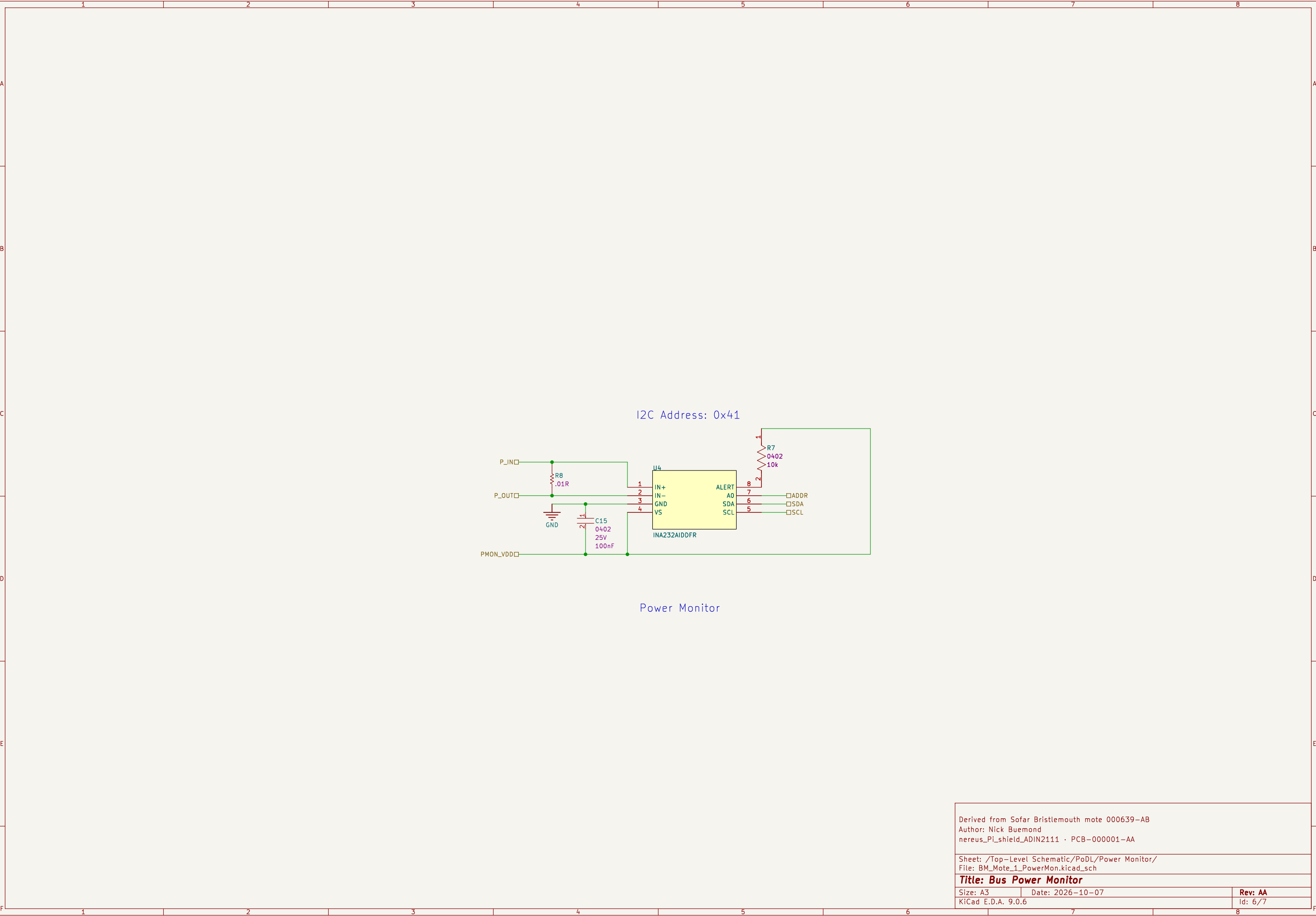
Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/PoDL/
File: BM_Mote_1_PoDL.kicad_sch

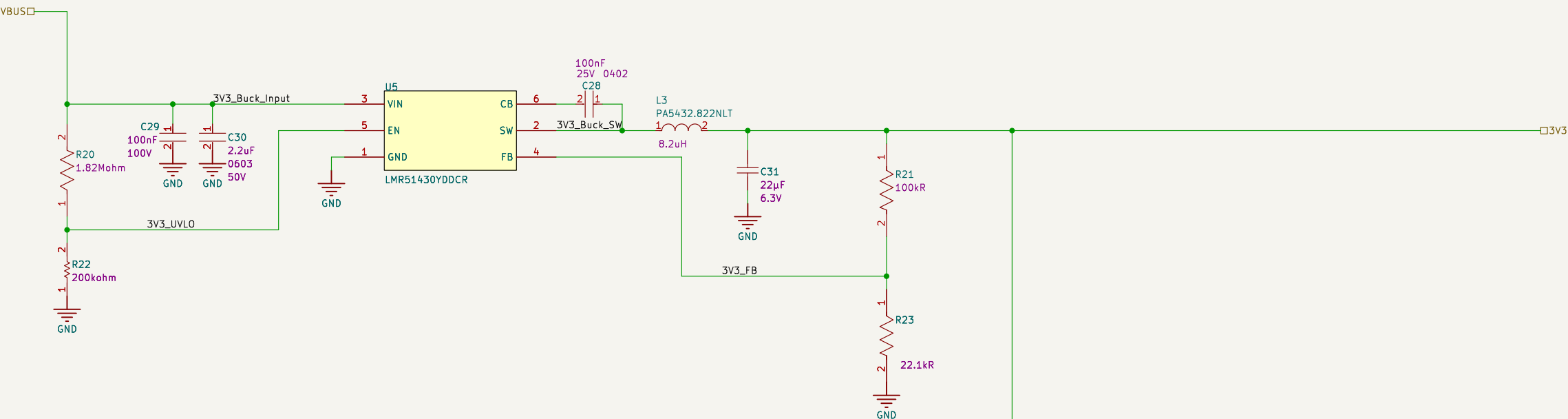
Title: PoDL (Power over Data Line)

Size: A3 Date: 2026-10-07
KiCad E.D.A. 9.0.6

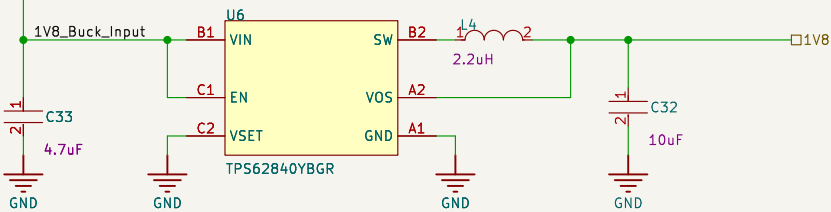
Rev: AA
Id: 5/7



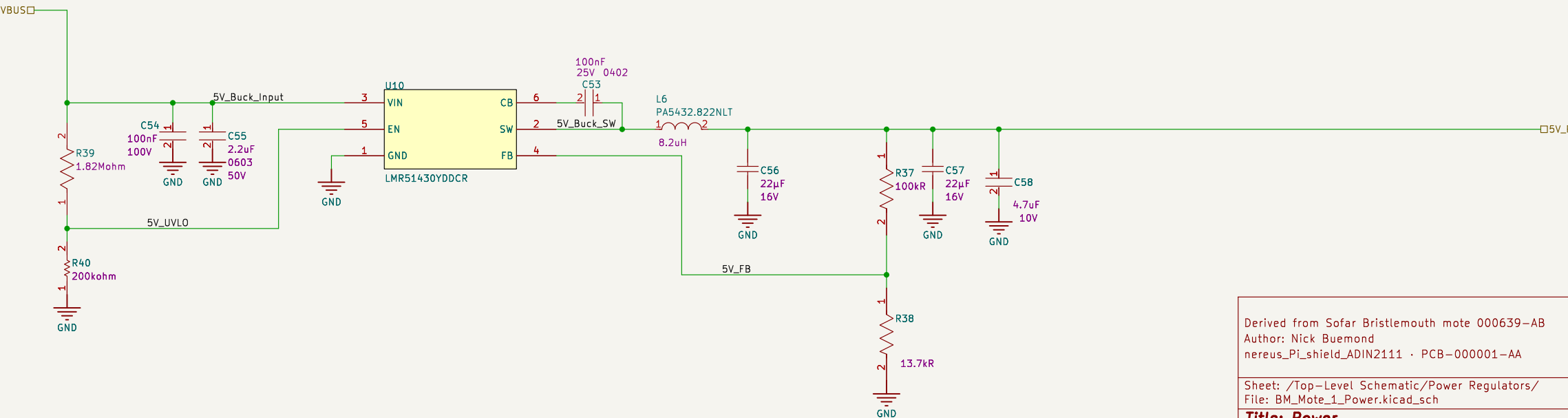
3V3 buck converter: U5 LMR51430, VBUS → 3V3



1V8 buck converter: U6 TPS62840, 3V3 → 1V8 (ADIN AVDD)



5V buck converter for the Pi: U10 LMR51430, VBUS → 5V_PI



Derived from Sofar Bristlemouth mote 000639-AB Author: Nick Buemond nereus_Pi_shield_ADIN2111 · PCB-000001-AA		
Sheet: /Top-Level Schematic/Power Regulators/ File: BM_Mote_1_Power.kicad_sch		
Title: Power		
Size: A3	Date: 2026-10-07	Rev: AA
KiCad E.D.A. 9.0.6		Id: 7/7